

Sean Joseph McIlwain

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About me

Data-driven bioinformatics professional with a passion for uncovering insights from data to solve data science problems. Skilled in R, with a strong background in statistical analysis. In my free time, I enjoy biking, hiking, and martial arts.

Work Experience

Facility Manager, University of Wisconsin-Madison

December 2019 - Present

- Managing the Cancer Informatics Shared Resource (CISR)
- Collaborate with multiple principal investigators to help analyze cancer related data
- Help collaborators publish papers and prepare grant proposal using bioinformatics

Assistant Scientist, University of Wisconsin-Madison

December 2015 - December 2019

- Analyzing RNA-Seq, ChIP-Seq, and proteomics data
- Applying pathway analysis tools to differential gene and protein expression
- TODO

Bioinformatics Researcher, Great Lakes Bioenergy Research Center, University of Wisconsin-Madison

October 2012 - December 2015

- Analyzed quantitative proteomic data from different samples
- Genome re-sequencing and annotation of microbial strains
- Developed scripts for analyzing chemical genomic datasets

Education

Doctor of Philosophy in Computer Science

University of Wisconsin - Madison
2003 - 2018

Masters of Science in Computer Science

University of Wisconsin - Madison 2001 - 2003

Bachelor of Science in Chemistry

Montana State University - Bozeman 1995 - 2000

Skills

- Data Analysis: Python, R
- Machine Learning: Torch
- Bioinformatics: DESeq2

Research and Publications

- M. Liu, Y. Zhu, **S. J. McIlwain**, H. Deng, A. R. Brasier, Y. Ge, M. E. Kimple, A. M. Baschnagel, “Characterizing plasma-based metabolomic signatures for metastasis in non-small cell lung cancer”, *Metabolites* 15(5):340, May 20, 2025, doi: 10.3390/metabo15050340
- J. D. Kratz, S. Rehman, K. A. Johnson, A. A. Gillette, A. Sunil, P. F. Favreau, C. A. Pasch, D. Miller, L. C. Zarling, A. H. Yeung, L. Clipson, S. J. Anderson, A. K. Steimle, C. M. Sprackling, K. K. Lemmon, D. E. Abbott, M. E. Burkard, M. F. Bassetti, J. C. Eickoff, E. F. Foley, C. P. Heise, R. J. Kimple, E. H. Lawson, N. K. LoConte, S. J. Lubner, D. L. Milkerin, K. A. Matkowskyj, C. B. Sanger, N. V. Uboha, **S. J. McIlwain**, I. M. Ong, E. H. Carchman, M. C. Skala, D. A. Deming, “Subclonal response heterogeneity to define cancer organoid therapeutic sensitivity”, *Sci Rep* 15(1):12072, April 9, 2025, doi: 10.1038/s41598-025-96204-2
- **S. J. McIlwain**, A. Hoefges, A. K. Erbe, P. M. Sondel, I. M. Ong, “Ranking antibody epitopes and proteins across samples from whole proteome tiled linear peptides”, *Bioinformatics* 5:btac637, Nov 5, 2024

List of Published Work in [MyBibliography](#)

Languages

- English: Fluent

Interests

- Hiking
- Biking
- Aikido
- Karate